

MANIFEST

BOB-095 — Go-side SSE-broker stress + chaos automation evidence

Revision: 1 Last modified: 2026-08-19T00:00:00Z

Scope

qBitTorrent-go/internal/service/sse_broker_stress_chaos_test.go — §11.4.85 stress + chaos coverage for SSEBroker (internal/service/sse_broker.go) under `go test -race`. Go-side sibling to BOB-094's Python tracker-fetch coverage.

Scenarios delivered (6 tests, all GREEN under -race)

STRESS

#	Test	Category
1	TestSSEBrokerStressChaos_SustainedSubscribeUnsubscribe	sustained-load — 1000 cycles + goroutine-leak guard concurrent — 100 subs ×
2	TestSSEBrokerStressChaos_ConcurrentSubscribers	5 msgs = 500

#	Test	Category
		delivered, no deadlock
3	TestSSEBrokerStressChaos_BoundaryDoubleUnsubscribe	boundary — double-close idempotency

CHAOS

#	Test	Category
4	TestSSEBrokerStressChaos_SlowSubscriberDoesNotBlock	backpressure isolation — default: clause proof
5	TestSSEBrokerStressChaos_UnsubscribeDuringPublish	RLock/Lock ordering under contention
6	TestSSEBrokerStressChaos_ProducerFloodBoundedByBuffer	producer- flood bounded by 10-msg buffer

§11.4.115 RED-first captures

Natural RED — §11.4.238 automated-QA-discovers defect

The initial test run of scenario #3 (double-unsubscribe) uncovered a REAL defect the source-side review had missed: `close(ch)` panicked with `close of closed channel`. Captured in evidence/
`red_capture_double_unsubscribe.txt`. Fix landed in the same commit:
`sse_broker.go` `Subscribe()` gates the `close` on registry membership so a second call is a safe no-op (§11.4.253 idempotency).

Synthetic RED — backpressure mutation

`sse_broker.go` `Publish()` was TEMPORARILY mutated by removing the `select { ... default: ... }` clause (turning it into a blocking send).
Scenario #4 wedged as expected; the `-timeout 10s` deadline triggered

and the test FAILED. Captured in evidence/
red_capture_backpressure.txt. The default: clause was restored
immediately after capture; the GREEN re-run is in evidence/
green_capture_after_restore.txt.

Evidence layout

- latency.json — p50/p95/p99 + goroutine deltas (scenario 1)
- concurrent.json — fan-out totals (scenario 2)
- boundary_double_unsubscribe.txt — safe no-op proof (scenario 3)
- chaos_slow_subscriber.json — publisher-not-wedged proof (scenario 4)
- chaos_unsubscribe_during_publish.json — race + panic counts (scenario 5)
- chaos_producer_flood.json — buffer-bounded delivery (scenario 6)
- red_capture_double_unsubscribe.txt — §11.4.115 RED (natural, defect-found)
- red_capture_backpressure.txt — §11.4.115 RED (mutation, restored)
- green_capture_after_restore.txt — final GREEN run, 6/6 PASS

Discipline citations

- §11.4.85 stress + chaos test types
- §11.4.115(F) machine-written verdict pairs — natural RED / mutation RED
- §11.4.161 rootless — no root/podman needed
- §11.4.14 cleanup on every exit — defer unsub() on every subscribe
- §11.4.3 SKIP-with-reason — evidence-dir creation failure, never fake pass
- §12/§12.6/§12.12 host safety — GOMAXPROCS=2 nice -n 19
- §11.4.238 automated QA discovers — scenario #3 caught a live defect
- §11.4.253 idempotency — the double-unsubscribe fix